SUMMARY · MK 6 BUSINESS ANALYTICS · WEEK 1

Introduction to Business Analytics

Business analytics uses data, statistics and models for better decisions. Four levels of analytics, DIKW, the 14-session plan, and analytics supporting rather than replacing judgement. Case: your syndicate's PT BetonPrima West Java turnaround.

Material

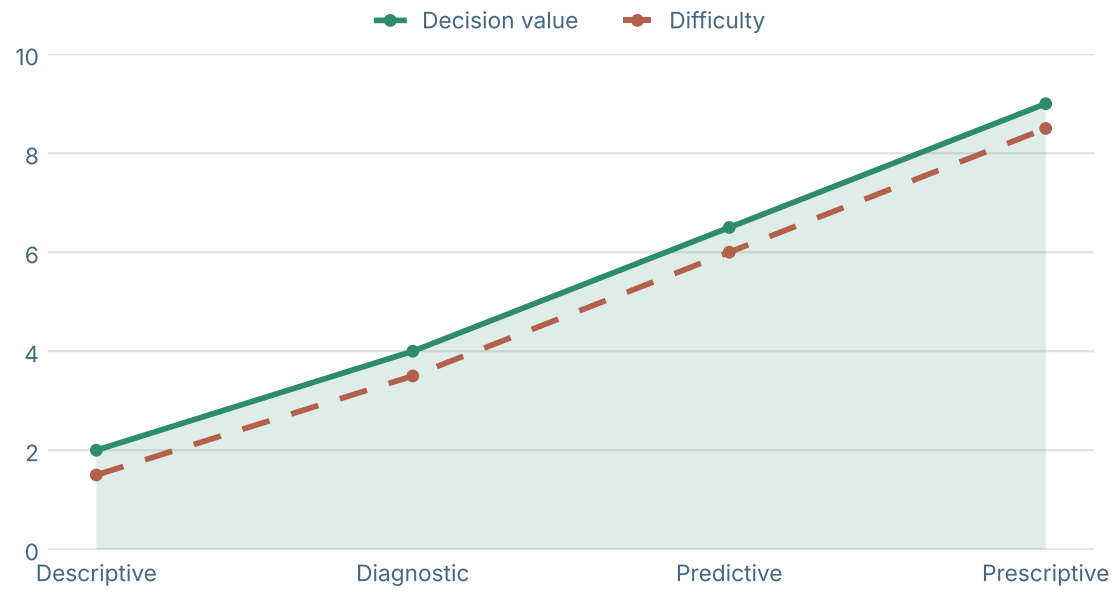
1 What business analytics is

Business analytics turns data into insight and action. Levels: descriptive, diagnostic, predictive, prescriptive — increasing difficulty and value. DIKW: Wamena temperatures → information → knowledge → wisdom (hold the festival in August).

2 Four levels of analytics: harder and more valuable

Deep dive

Value vs difficulty by level (conceptual)





3 Working principles

Start from the decision; expect messy data and conflicting definitions; analytics informs judgement; correlation is not causation; report ranges.

4 BetonPrima in charts: four levels in one case

Deep dive

−20%	214	5.4 days

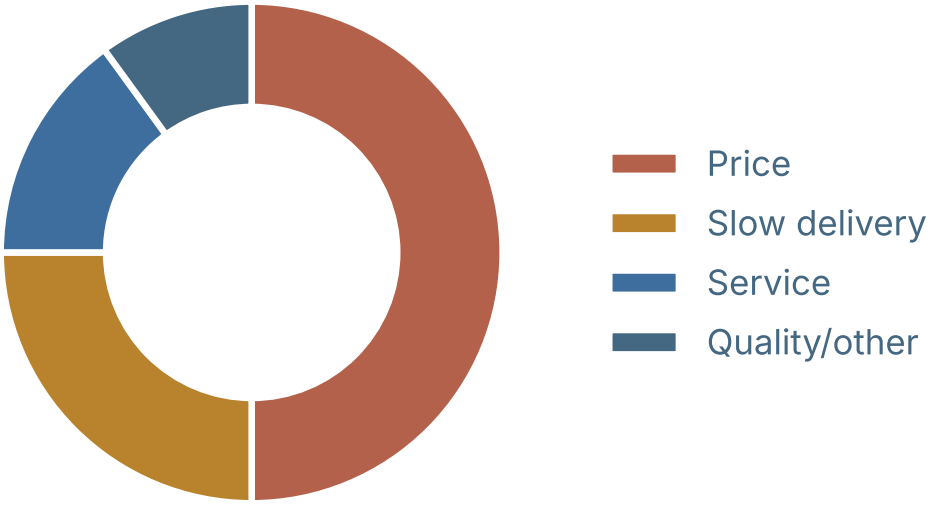
West Java volume

Accounts lost

Order-to-site

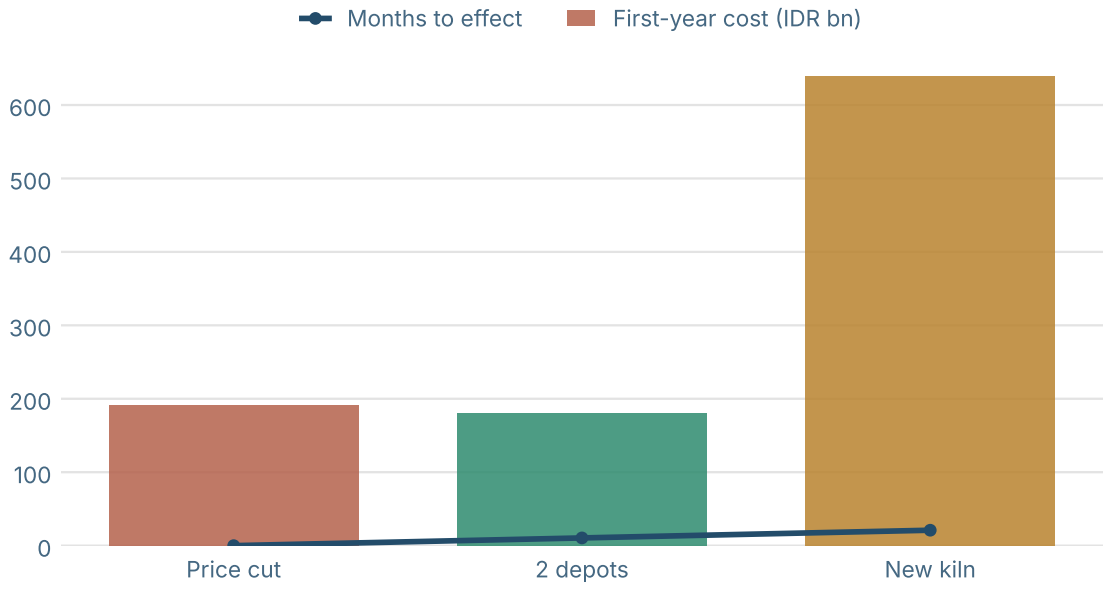
91% → 68%
On-time

Churn reasons in the system log



Interviews showed many 'price' entries were really 'late when raining'.

Three options: first-year cost and months to effect



Quick maths

Discounting at 48bn/quarter ≈ 192bn a year, more than two depots (180bn).

Key takeaways

1. Four levels.
2. DIKW.
3. Start from the decision.
4. Analytics supports judgement.
5. Mislabelled data hides root causes.

Case studies

Case & worked answers

Your syndicate: PT BetonPrima West Java turnaround

Descriptive: West Java volume fell 20% while demand was flat; 214 accounts closed. Diagnostic: churn logged as "price" was really rain-related late delivery (rigid drop-down — a capture problem); lead time 5.4 days vs 3.0, on-time 91% → 68%; weather and sales neglect ruled out. Prescriptive: two depots (IDR 180bn) beat price cuts and a kiln rebuild. Add: fix the capture form and compute depot payback.

Practice questions

Try answering first. The answer key is below each question.

1. 'What should we do?' is...

A. Descriptive
B. Diagnostic
C. Predictive
D. Prescriptive

Answer: D. Output is a decision.

2. BetonPrima's misclassified reasons are a defect at…

A. Pipeline
B. Capture (the form)
C. Dashboard
D. Model

Answer: B. A rigid single-choice drop-down.

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3. The option that attacks the root cause…

A. Price cut
B. Two depots
C. New kiln
D. More sales staff

Answer: B. Root = logistics bottleneck.

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4. 'August in Wamena tends to be warm' is…

A. Data
B. Information
C. Knowledge
D. Wisdom

Answer: C. A general pattern.

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5. Analytics should start from…

A. Available data
B. The decision to be made
C. The newest tool
D. A dashboard

Answer: B. What, when, what changes.

Sources

- WEEK 1 MK06 P3MD (slide berupa gambar)
- Course plan MK6
- PT BetonPrima Analytics — Syndicate 3 Kelas E (tugas kelompokmu)